

RAJ KRISHNAN VIJAYARAJ

+1 (249) 876-0601 • Toronto, Canada • rajkrishnanv@gmail.com • [LinkedIn](#) • [GitHub](#) • unni12345.github.io

EXPERIENCE

Machine Learning Engineer (R&D)

Nov 2025 – Present

Sanofi Vaccines Canada, Toronto, Canada

- Cut model deployment time from 4 weeks to 2 days with CI/CD pipelines shipping 20 bioinformatics models to AWS HealthOmics across DEV/UAT/PROD, now used by 100+ research scientists.
- Built the compliant multi-account infrastructure underneath (ECS Fargate, IAM/OIDC federation, SQS, S3, Lambda) with least-privilege promotion gates, now the reference architecture across Sanofi's R&D platform.
- Project lead and technical point of contact, directing 3 engineers; own production reliability (retry/DLQ/status propagation, CloudWatch metrics and alarms) and nanobody workflow orchestration.
- Building an agentic auto-deploy layer that drives the framework end to end, removing the remaining manual promotion steps (in progress).

Machine Learning Engineer

Jul 2024 – Oct 2025

Appfabs (contracted to LexisNexis), Toronto, Canada

- Shipped production Text-to-SQL and RAG agents (Spark, LangChain, Hugging Face) with distributed multi-node inference on KubeRay and Docker, released through Jenkins.
- Cut end-to-end query latency by 50% by reworking data processing pipelines, schema indexing, and high-throughput GraphQL/REST API gateways.

Applied Research Intern

May 2024 – Jul 2024

Vector Institute, Toronto, Canada

- Fine-tuned LLaMA-3.1 across multi-GPU clusters with FSDP and served it behind low-latency vLLM endpoints.
- Designed a diagnostic RAG system whose chain-of-verification step held hallucinated clinical entities under 1%.

Machine Learning Engineer (Mitacs Fellow)

Sep 2022 – May 2024

Scribble Data, Toronto, Canada (*concurrent with MEng*)

- Benchmarked LoRA against INT8/INT4 quantization, cutting inference memory overhead at minimal accuracy loss and making edge deployment viable.
- Built privacy-aware LLM agents for legal document analysis with domain masking and automated PII redaction.

Backend Developer

Jul 2019 – Nov 2021

G2, Bengaluru, India

- Built high-throughput GraphQL APIs and async ETL (Rails, PostgreSQL, Sidekiq); cut batch data-export runtime from 72 hours to under 20.

EDUCATION

University of Toronto — MEng, Electrical & Computer Engineering (GPA: 3.94/4.00)

2022 – 2023

NIT Tiruchirappalli — B.Tech, Electrical and Electronics Engineering

2015 – 2019

PUBLICATION

R. K. Vijayaraj. *Embeddings to Diagnosis: Latent Fragility under Agentic Perturbations in Clinical LLMs.* Workshop on Evaluation of Generative AI, KDD 2025. [arXiv:2507.21188](https://arxiv.org/abs/2507.21188)

SELECTED PROJECTS

- **Eval Harness (SPAR)** — Score-based robustness metrics for computer-use agents under prompt injection.
- **Multimodal Clinical RAG** — GPT-4V + ClinicalBERT retrieval with chain-of-verification grounding.
- **Diagnostic LLM Audit** — Diagnostic LLMs fine-tuned on synthetic notes; perturbation-audited for stability.

SKILLS

ML & GenAI: PyTorch, Hugging Face, LangChain, RAG, LoRA/PEFT, INT8/INT4 quantization, FSDP, vLLM

Evaluation & Reliability: LLM evals, perturbation/robustness auditing, adversarial red-teaming, grounding

Systems & Cloud: AWS (HealthOmics, ECS, SQS, S3, Lambda), Docker, Kubernetes/KubeRay, Terraform, Spark, CI/CD

Languages & Data: Python, SQL, C++, Ruby, Snowflake, GraphQL, REST, PostgreSQL